

Part 1: Theoretical Analysis (40%)

Q1: Edge AI – Latency and Privacy

Question: Explain how Edge AI reduces latency and enhances privacy compared to cloud-based AI. Provide a real-world example (e.g., autonomous drones).

Answer:

Edge AI refers to the deployment of artificial intelligence algorithms directly on local devices (e.g., smartphones, embedded systems, drones) rather than relying on centralized cloud infrastructure. This approach significantly reduces **latency** by enabling real-time data processing at the source, eliminating the round-trip time required to transmit data to and from the cloud. For instance, in **autonomous drones**, object detection and navigation decisions must occur instantly; processing this data locally ensures real-time responsiveness and avoids delays that could lead to mission failure or safety risks.

Moreover, Edge AI enhances **privacy** by minimizing data transmission. Sensitive data (e.g., facial images or biometric data) is processed locally rather than uploaded to external servers, reducing the risk of interception or misuse. This is particularly important in applications such as **health monitoring wearables** or **smart surveillance systems**, where privacy regulations like GDPR apply.

Q2: Quantum AI vs Classical AI in Optimization

Question: Compare Quantum AI and classical AI in solving optimization problems. What industries could benefit most from Quantum AI?

Answer:

Classical AI relies on conventional binary computation, which can struggle with highly complex optimization problems due to the exponential growth of possible solutions. In contrast, **Quantum AI** leverages quantum computing principles—such as **superposition** and **entanglement**—to explore multiple possibilities simultaneously, potentially solving certain optimization problems exponentially faster.

Quantum AI shows promise in fields requiring **combinatorial optimization**:

- **Pharmaceuticals:** Accelerated drug discovery by simulating molecular interactions more efficiently.
- **Finance:** Portfolio optimization and risk assessment based on massive, dynamic data sets.

- **Logistics:** Solving vehicle routing problems in real-time to enhance delivery efficiency.
- **Energy:** Optimizing power grid distribution to balance demand and reduce waste.

While still in early development, Quantum AI represents a disruptive force for industries that depend on rapid, high-dimensional optimization.

Q3: Human-AI Collaboration in Healthcare

Question: Discuss the societal impact of Human-AI collaboration in healthcare. How might it transform roles like radiologists or nurses?

Answer:

Human-AI collaboration in healthcare represents a **synergistic model**, where AI systems augment—not replace—healthcare professionals. For example, AI-powered imaging tools can assist **radiologists** by rapidly identifying anomalies such as tumors or fractures in scans, reducing diagnostic errors and expediting patient care. Instead of manually examining thousands of images, radiologists can focus on validating AI results and making informed treatment decisions.

Similarly, **nurses** can benefit from AI systems that monitor patient vitals in real time and alert them to early signs of deterioration, such as sepsis. This reduces response times and improves patient outcomes.

Societally, this collaboration enhances the **efficiency and quality of care**, addresses workforce shortages, and allows healthcare professionals to dedicate more time to empathy-driven aspects of care—like communication and emotional support. However, it also demands **retraining and trust-building**, ensuring that medical workers feel empowered rather than replaced by AI.